

Processors will have more processing power than the human brain

5G has not been formally defined yet, but it has become imperative to connect machines and people to each other across the planet. However, will it be just another increase to the peak bandwidth? Rahman Jamal, global technology and marketing director at National Instruments, explains as he speaks with Dilin Anand from EY



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Q. From the technological perspective, what is driving the Internet of Things (IoT)?

A. There are multiple technologies driving this trend. There are laws that make these things affordable, such as Moore's Law that states that the number of transistors per unit area on integrated circuits doubles every 18 to 24 months. Other driving elements are Metcalfe's Law, Nielsen's Law, an increasing number of systems from wired to wireless and, finally, sensors that are the sense organs of the IoT.

Q. How are these laws and technologies driving the IoT?

A. Moore's Law has guided us over the past few decades as we have made processors smaller, faster and more affordable, putting a sea of transistors on a tiny chip. In the coming decades, technologists estimate that processors will have more power than the human brain.

Metcalfe's Law has shown us that every time we add a node to a network, the value increases by a factor of two, encouraging us to connect everything and synthesise data.

Nielsen's Law is about the bandwidth of the data highway, namely, the Internet, and states that a high-end user's connection speed grows by 50 per cent every year.

The battery technology needs to improve continually with new compounds and enabled electronics to become detached from the grid for longer amounts of time.

With sensors, the more sensing you add to things that are connected, the more context-aware these become. Combined with data analytics, these can give more valuable information.

Q. How important will the 5G telecommunications networks be to the success of the IoT?

A. Though today's 4G networks continue to offer faster data access, the road beyond LTE and LTE-A is far from clear.

5G networks must accommodate many more users and devices while delivering more data to each user at any instant. Researchers envision not only a 5G network with unprecedented data rates and mobile access but also an opportunity to redefine the network to accommodate a wealth of new and diverse connected devices.

5G targets peak data rates per user in the range of

10Gbps (over 1000 times 4G). To provide a frame of reference, a user can download an HD video in seconds with 5G, while it took about 40 minutes to download the same video using the highest speed networks in good conditions.

Q. What kind of 5G research is happening right now?

A. There are many universities and research institutes that are looking at different aspects of 5G. For instance, the 5G wireless lab in Germany, Technische Universität Dresden, led by Dr Gerhard Fettweis, is looking at 5G from the waveform perspective and designing orthogonal frequency division multiplexing (OFDM) prototyping systems based on NI's RF and communication tools.

At New York University, Dr Theodore S. Rappaport is looking into millimetre waves around 28GHz, 38GHz and the 71GHz to 76GHz range.

Then, Nokia Networks has prototyped a mmWave communication link and is achieving data rates 100 times above current 4G rates with deterministic latency. By using NI's integrated hardware and software baseband platform, Nokia plans to demonstrate the viability of high-frequency millimetre wave as an option for 5G radio access technology.

Q. How is technology taking this to the next level?

A. Let us take an example from India. The traffic here seems to regulate itself. There are lanes but no one follows these. Roads keep changing from one-lane to multiple-lanes, and back. This happens because there is this collective intelligence of the people operating the vehicles.

In contrast, in the west, we find a more centrally-controlled traffic system. There is no flexibility because the lanes are of fixed size and the whole bandwidth is not used all the time; you might find one side of a four-lane highway completely empty, while the other side is jammed. In India, the whole bandwidth gets regulated.

Q. Will the next generation of connected cars need 5G?

A. When cars begin communicating with each other, these would not just be sending single data; these would also send waveform data, which requires a different bandwidth when these cars really need to talk to each other, and 5G will hopefully meet these needs. ●